I/B/E/S Earnings Data: Cleaning and Surprise Construction

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> *This report covers 287,844 quarterly earnings announcements (1995-01-03 to 2014-06-30) cleaned from raw I/B/E/S analyst-estimate and CRSP share-price data, and how each announcement's earnings surprise was computed while guarding against look-ahead bias, stock-split artifacts, and ambiguous company identifiers.*

What an "earnings surprise" is

Every quarter, public companies report their actual earnings per share. Wall Street analysts publish estimates of that number beforehand, and the stock market's reaction depends mostly on the GAP between what was reported and what was expected, not the raw number itself. This project calls that gap the "earnings surprise." This report covers a cleaned table of 287,844 such quarterly announcements, spanning 1995-01-03 to 2014-06-30, built from raw I/B/E/S (Institutional Brokers' Estimate System) analyst-estimate data and CRSP share-price data.

Why timing matters: point-in-time discipline

Analyst estimates keep getting updated for years after a company reports, as analysts revise their forecasts for later quarters. That creates a trap: it is easy to accidentally compute a company's "surprise" using an estimate that was published AFTER the announcement, which means the estimate could already reflect the real result. Testing whether a market reacted to information it never actually had would make any later finding meaningless. So every estimate used here is checked to have been published strictly before the announcement it is compared against. Of all announcements, 10,916 individual analyst estimates were found to have been published on or after the announcement date and were excluded for exactly this reason -- an expected, mechanical filter, not a data error.

Two ways to measure a surprise

This project computes surprise two different ways. The preferred measure, `sue_analyst`, compares the actual result to the average Wall Street analyst estimate for that quarter, scaled by how much analysts disagreed with each other. It needs real analyst coverage to exist. A mechanical fallback, `sue_seasonal`, does not: it compares the result to the same quarter a year earlier, scaled by how volatile that year-over-year change has historically been for that company. Across the full sample, a real analyst estimate was available for 67.5% of

announcements and the mechanical measure was computable for 76.8% (the two overlap, since both can exist for the same announcement). This project prefers the real analyst estimate whenever it exists and only falls back to the mechanical measure when it does not, which is why the mechanical measure ends up actually USED for a smaller 20.4% of announcements, and 12.1% of announcements have neither and are kept in the data with no surprise value rather than being dropped. Coverage is far from constant over time -- the chart below shows which measure was available in each year of the sample.

How available coverage shifted, 1995-2014

The chart below shows, for each announcement year, what share of that year's earnings reports got a real analyst-based surprise, what share fell back to the mechanical measure, and what share had neither. Analyst coverage is thinner in the earliest years of the sample and grows steadily through the 2000s and 2010s as more companies gained regular analyst coverage. The very start (1995-1996) and very end (2014) of the window are both visibly thinner than the interior years: 1996 sits at the edge of the one-year lookback the mechanical measure needs, and 2014 falls just past a deliberate cutoff in how far the underlying analyst-estimate data was pulled -- both are known edge effects of the sample's boundaries, not data-quality problems.

What the surprise measure looks like across all events

Most earnings announcements are close to what analysts expected -- a surprise near zero -- with progressively fewer large beats or misses the further from zero you go. That shape (a peak near zero, thinning tails) is exactly what the chart below shows for the combined surprise measure used across this project, `sue_primary` (the real analyst-based measure where available, the mechanical fallback otherwise).

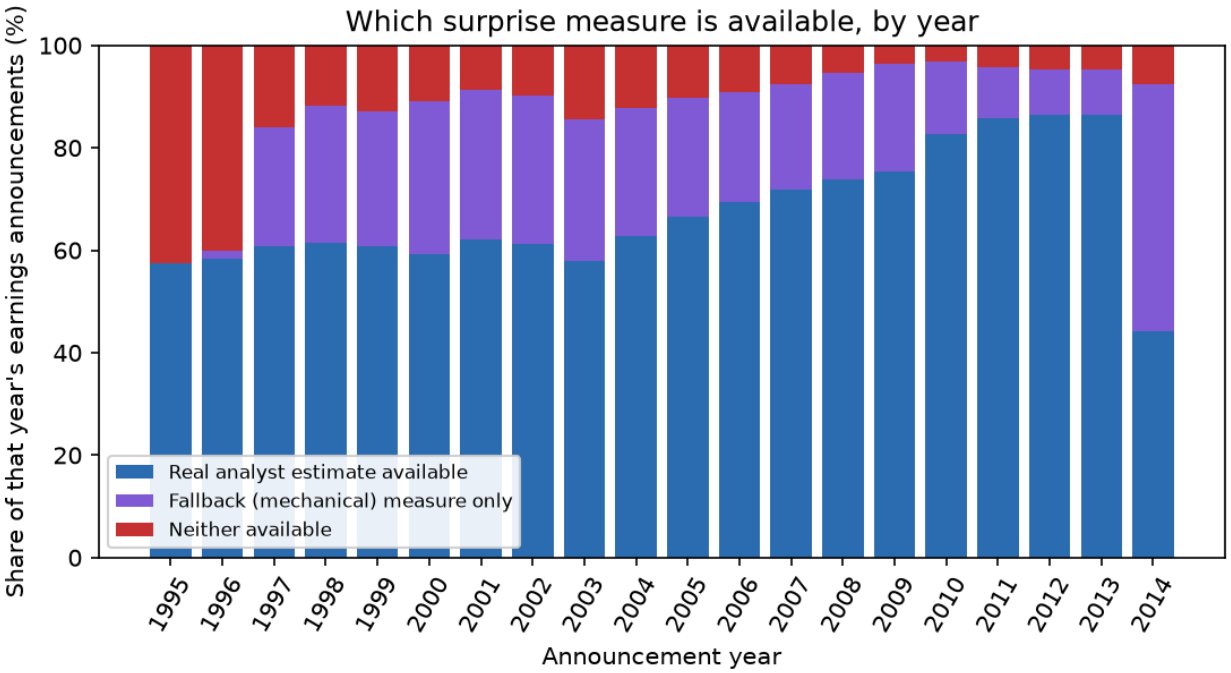
Data-quality handling: which company, and on what basis

Two separate identification problems had to be handled honestly rather than guessed. First, a small share of announcements could not be tied to a single, unambiguous company identifier: 7.5% of announcements carry more than one candidate company ID, most often because the company was reassigned a new internal ID after a corporate restructuring. These are flagged in the data rather than silently resolved one way or another. Second, stock splits change how many shares a company's earnings are divided by, which can make a perfectly normal result look like a huge surprise if the reported figure and the analyst estimate are not on the same per-share basis. For 689 announcements where a split fell in a window where this mattered, the reported figure was independently confirmed correct; for a further 3,090, the evidence was not strong enough to confirm either way, so the surprise measure is left blank for those rows rather than guessed. The chart below shows the full breakdown.

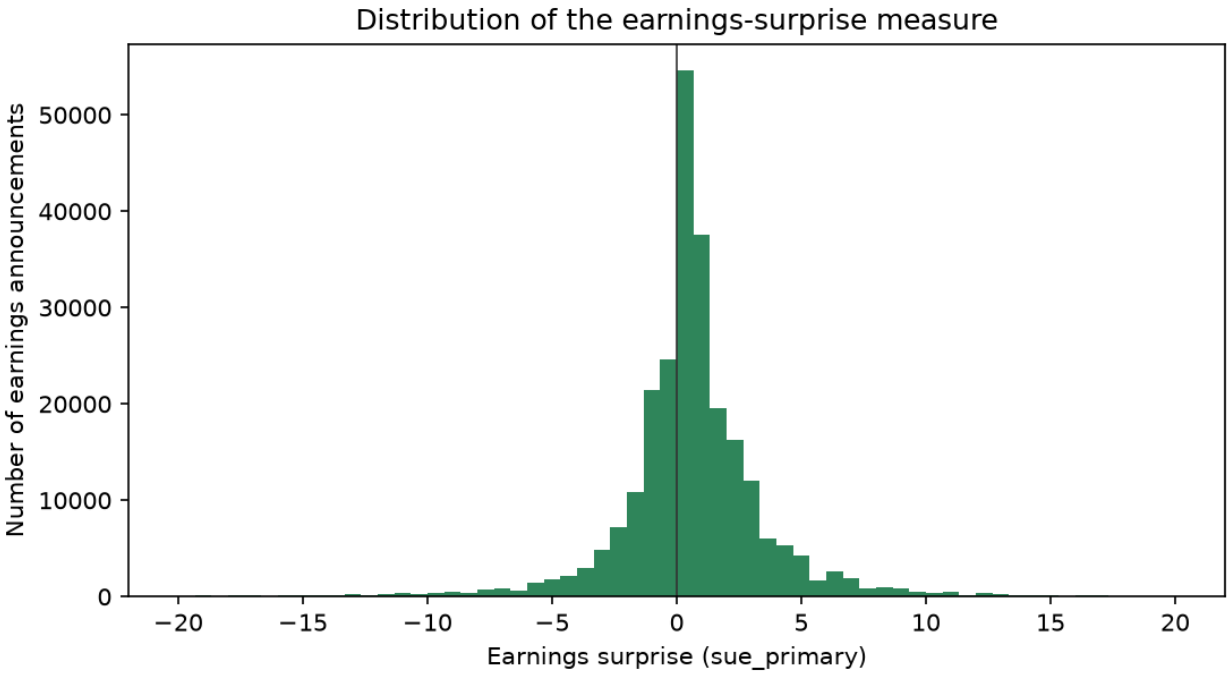
What this means for the project

This slice turns raw analyst-estimate and earnings-actuals data into one clean number per company per quarter that says how much that quarter's result beat or missed expectations,

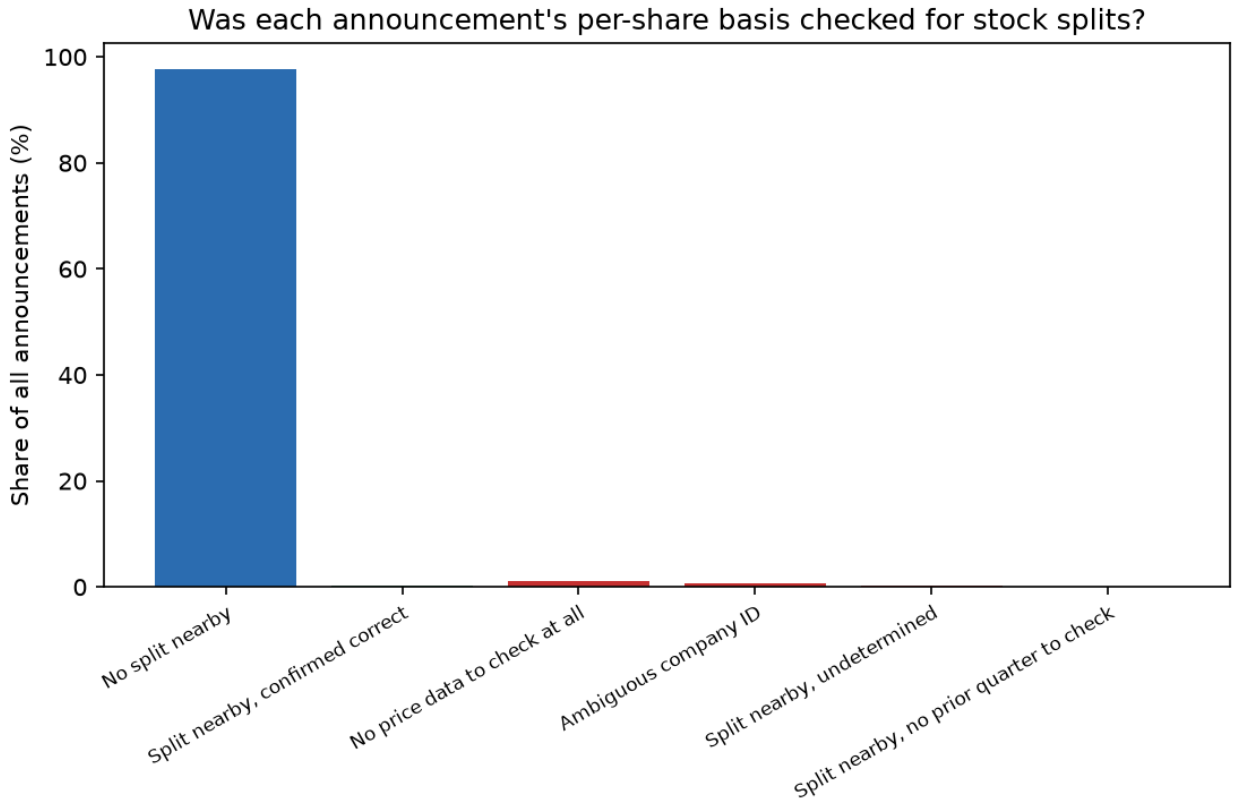
with three specific traps -- using information not yet public, mismatched company identifiers, and stock-split artifacts -- checked and either corrected or honestly flagged rather than silently ignored. That is the input the rest of this project needs: later phases will match each of these announcements to the company's actual stock-price behavior in the days and weeks that follow, to test whether the market keeps drifting in the direction of the surprise instead of reacting all at once.



Real analyst-based coverage (blue) grows from roughly half of announcements in the mid-1990s to over 85% by the early 2010s; the mechanical fallback (purple) fills much of the remaining gap in most years. The share with neither measure (red) is highest at the start of the sample and shrinks steadily.



Distribution of the pooled earnings-surprise measure across 253,131 announcements that have one. Values are clipped to -20 to +20 for readability; 0.8% of announcements have a more extreme surprise than that and are not shown on this chart, but are still included in the underlying data. Most announcements cluster close to zero, meaning the actual result was close to what analysts expected.



For the large majority of announcements, no stock split fell close enough to the announcement to raise a per-share-basis question at all (blue). Where one did, the reported figure was independently confirmed correct far more often than not (green); the remaining, smaller categories (red) are cases left honestly unresolved -- either a real gap in the underlying price data, an ambiguous company identifier, or evidence too weak to confirm which basis applies -- rather than guessed.